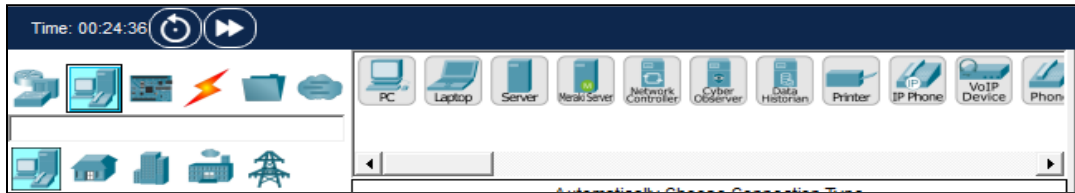


Practical No: 4

AIM: Using Packet Tracer ,Create a basic Network of One Server and two computers and two mobile/movable device using appropriate network wire,and verify connectivity

STEP 1

Open Cisco Packet Tracer.



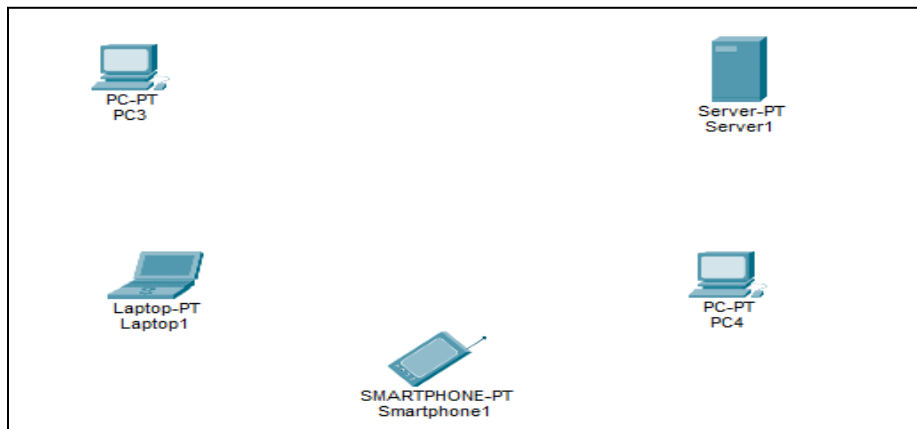
STEP 2

From End Devices,place:

2PCs

1 Server

2Mobile Devices(Laptop/Smartphone)



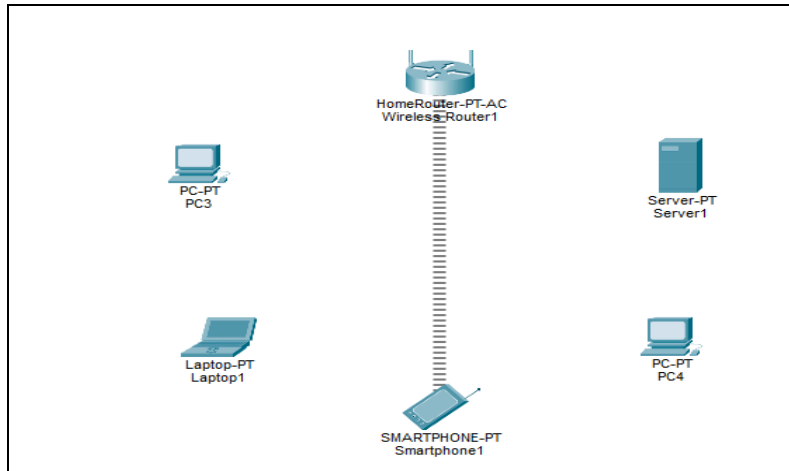
STEP 3

From Network Devices → Wireless Devices,Select Home Router



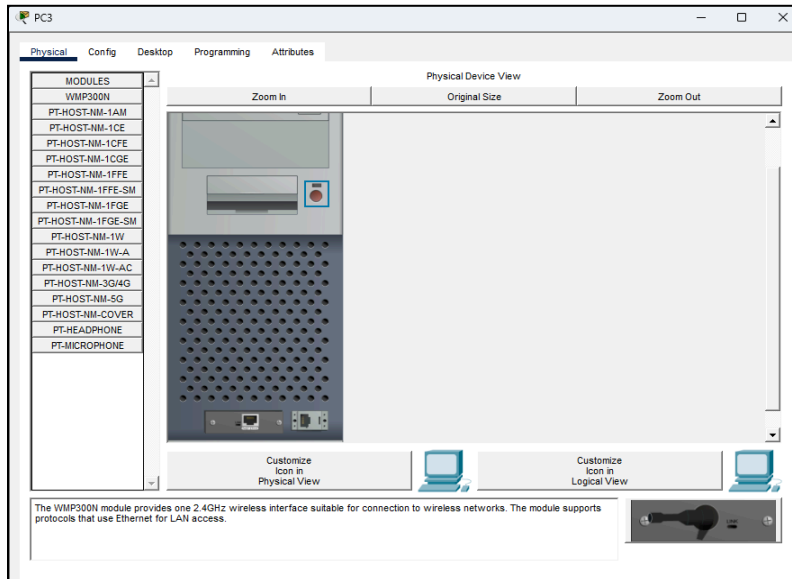
STEP 4

Drag it onto the workspace.



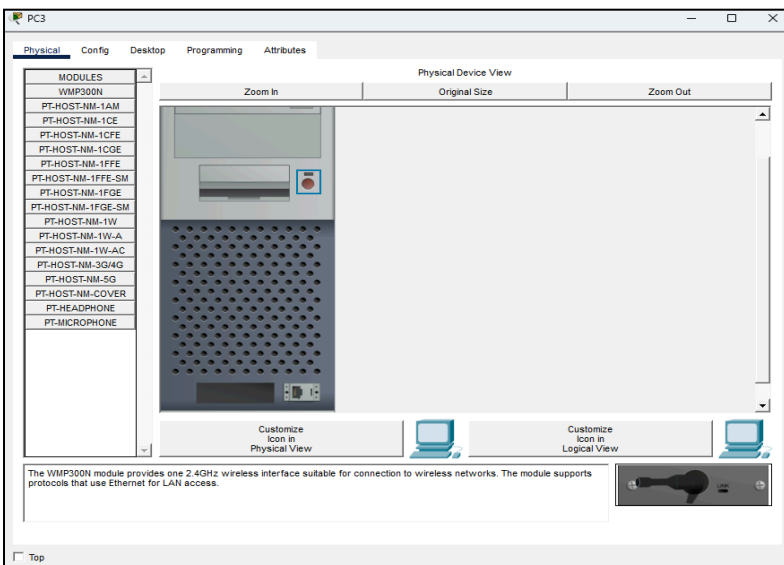
STEP 5

To enable wireless connectivity click on PC → Physical tab, turn OFF the PC



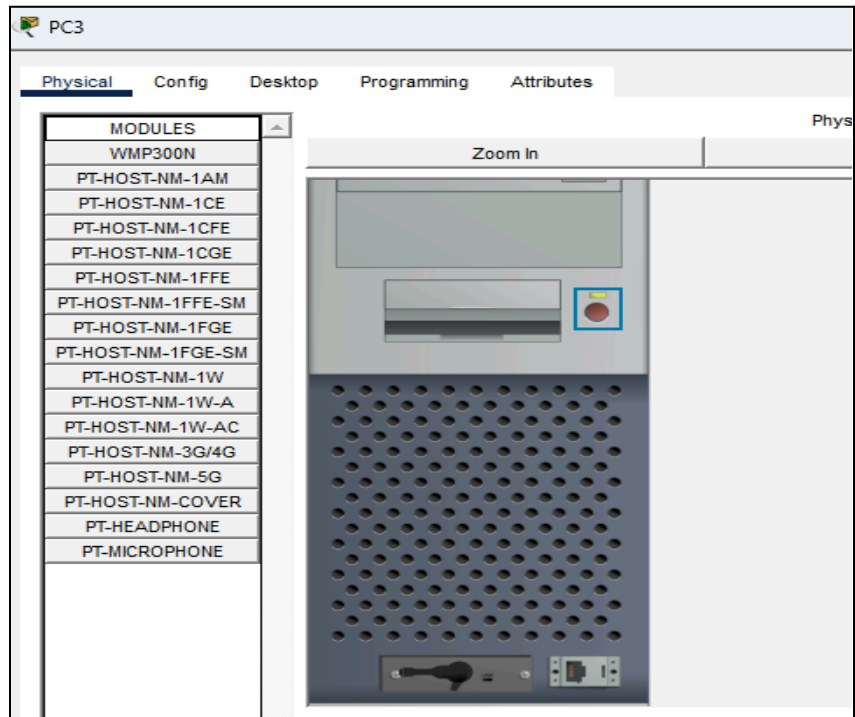
STEP 6

Remove the Ethernet module, insert the Wireless module (WMP300N)



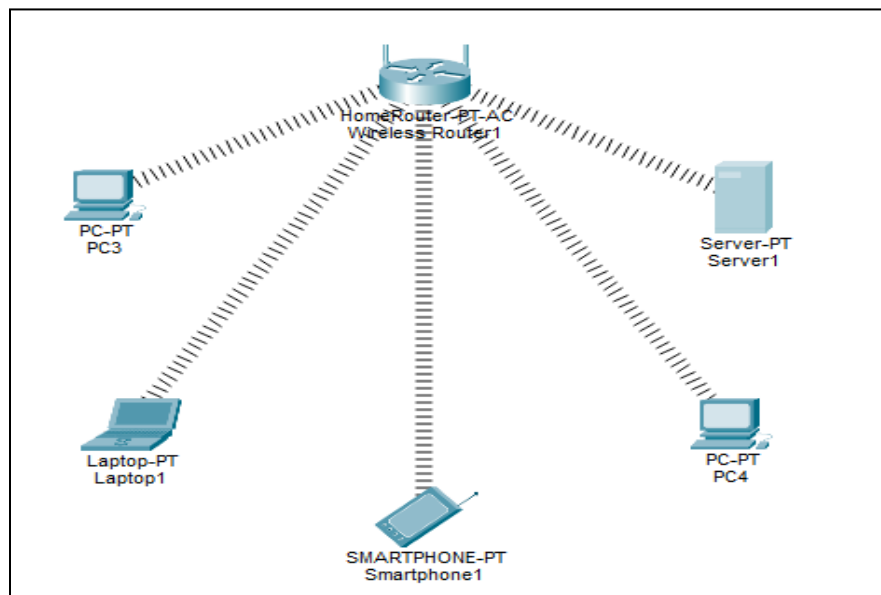
STEP 7

Now Turn the PC On

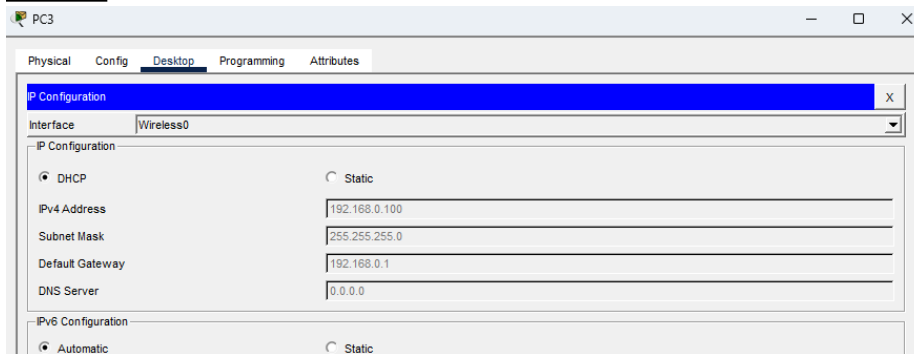


STEP 8

Repeat the same steps to all end devices.

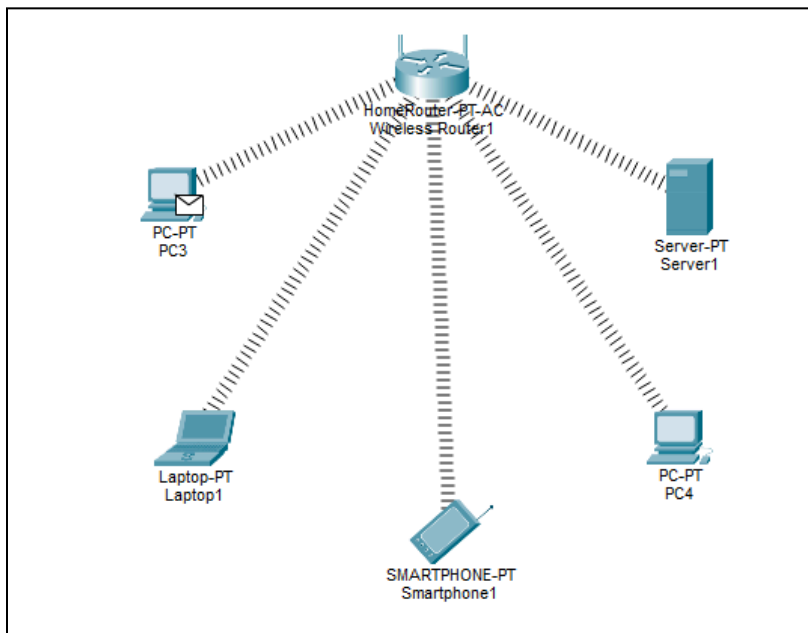


STEP 9



STEP 10

Click on the Envelope icon from the right-side toolbar. Then click on PC3 → PC4



STEP 11

Successful replies confirm that the network is working properly.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	PC4	ICMP		0.000	N	0	(e...	

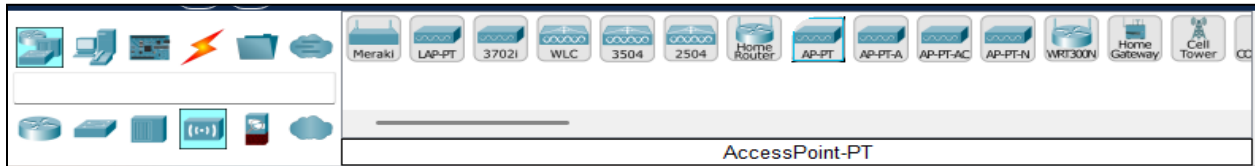
Practical No: 5

AIM: Using Packet Tracer, Create a wireless network of multiple PC's using appropriate access point

Step 1

Open Cisco Packet Tracer

From the **Network Devices** section, select **Wireless Devices** and choose a **Wireless Access Point**.



Step 2

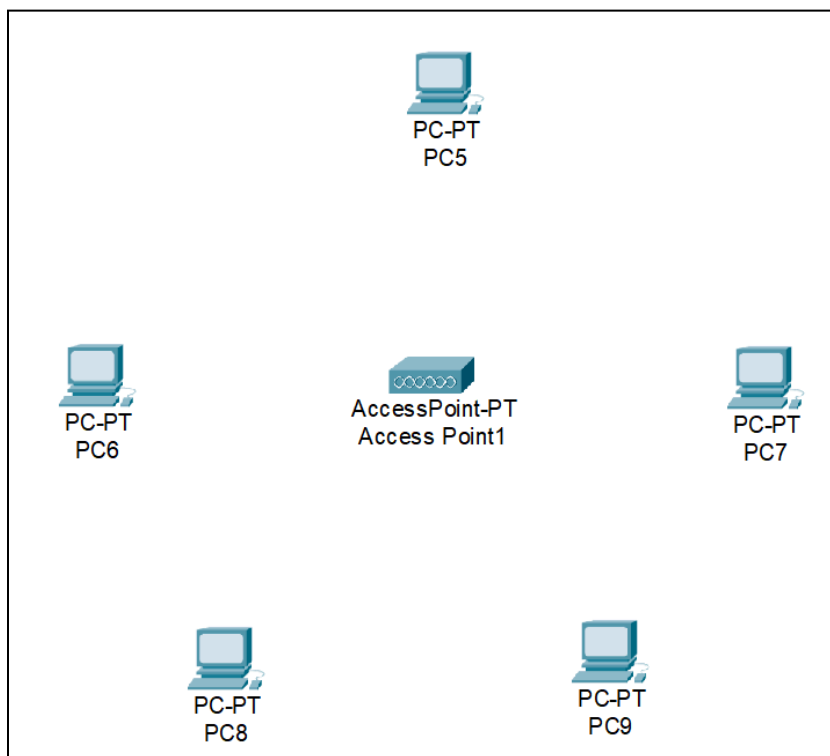
Place the **Wireless Access Point** in the workspace.

From **End Devices**, select multiple **PCs**



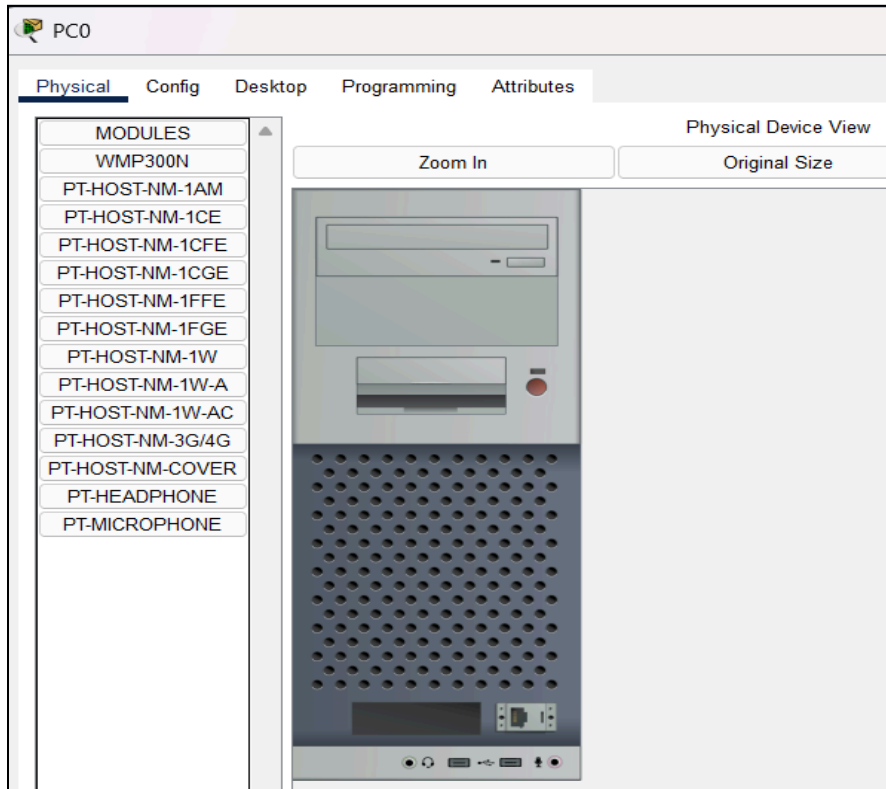
Step 3

Place them around the wireless access point.



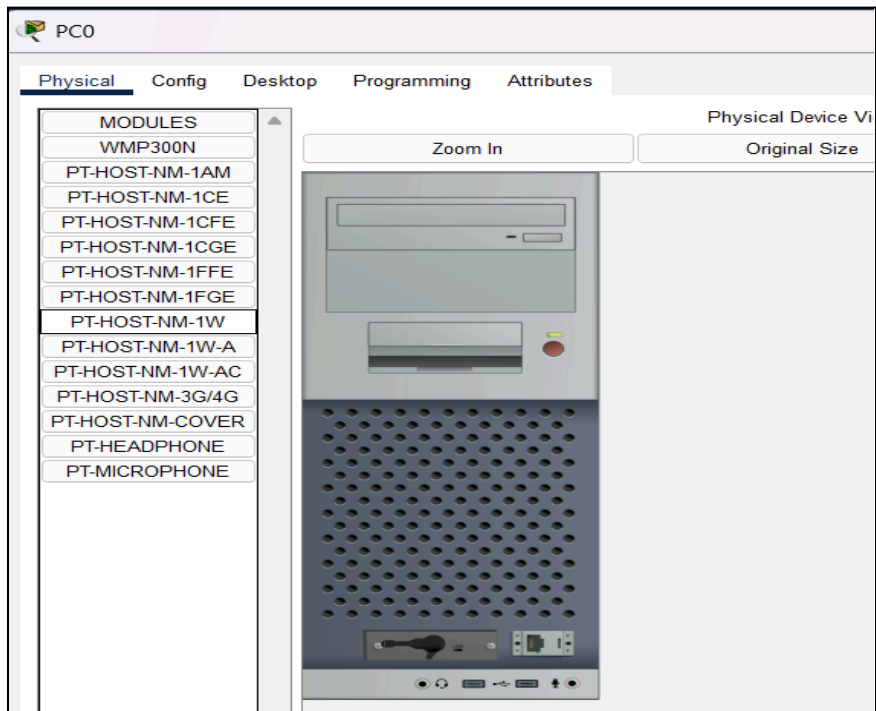
Step 4

Click on the PC → **Physical tab** → turn OFF the PC. Remove the Ethernet module



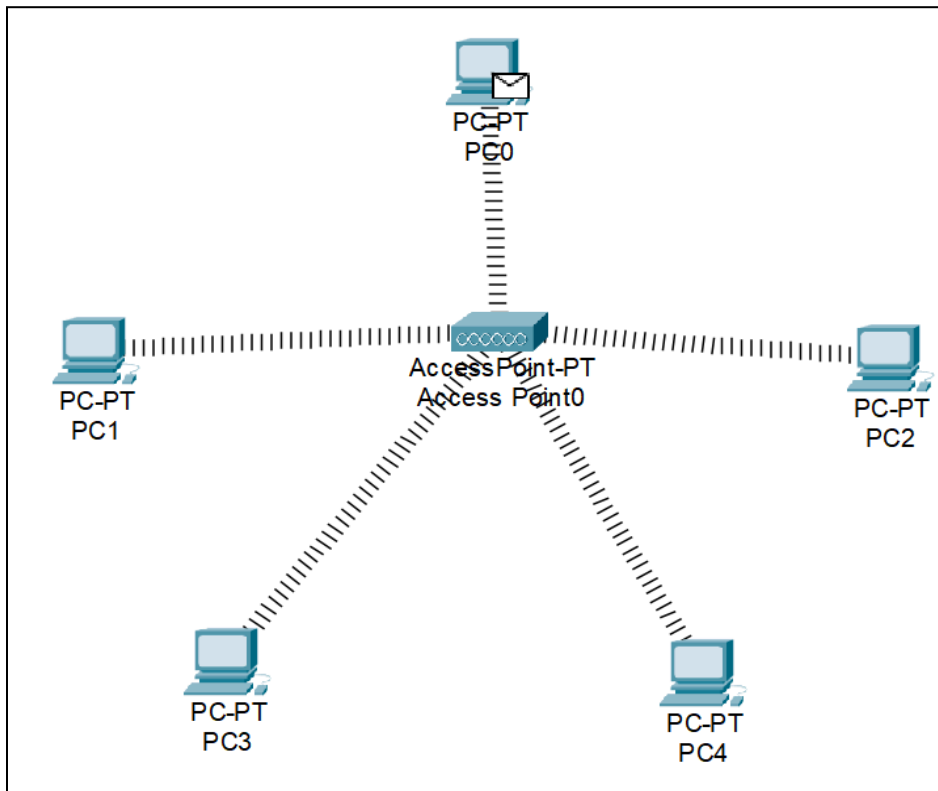
Step 5

Select **PT-HOST-NM-1W** from the Modules tab, and then turn the PC ON



Step 6

Now test the connection by sending packets.



Step 7

Successful replies confirm that the PCs are connected to the wireless access point and the network is working properly.

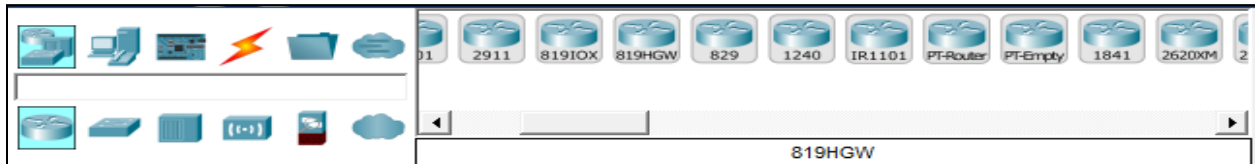
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC3	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC4	ICMP		0.000	N	2	(edit)	(delete)

Practical No: 6

AIM: Using Packet Tracer, Create a network with three routers with OSPF and each router associated network will have minimum three PC, Show Connectivity

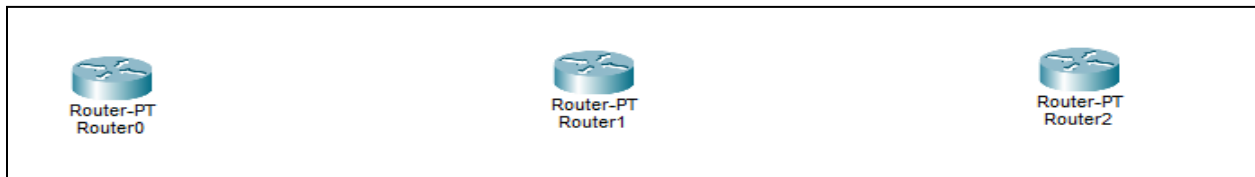
Step 1

Open Cisco Packet Tracer.



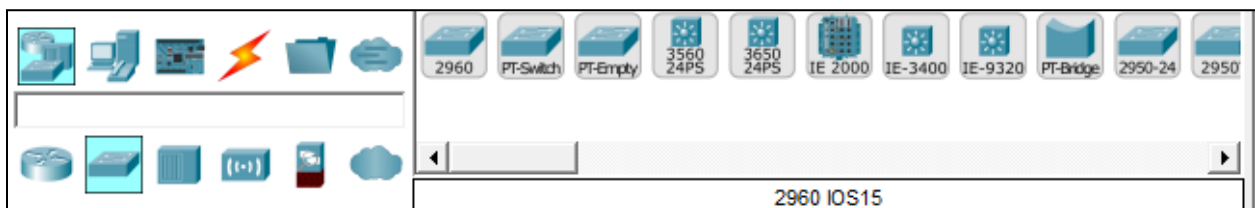
Step 2

From Network Devices → Routers, drag three routers (Router1, Router2, Router3) into the workspace.



Step 3

From Network Devices → Switches



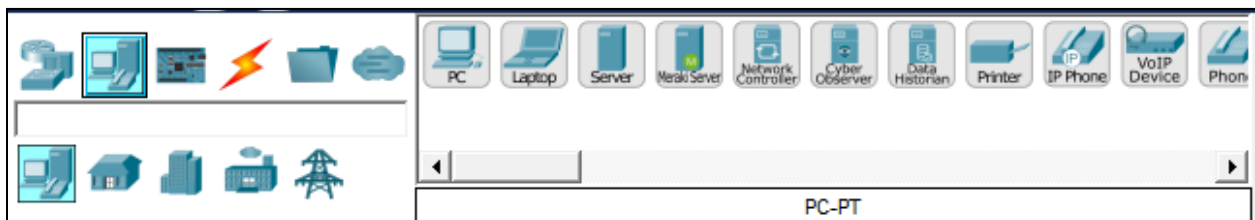
Step 4

Place **three switches**, one near each router.



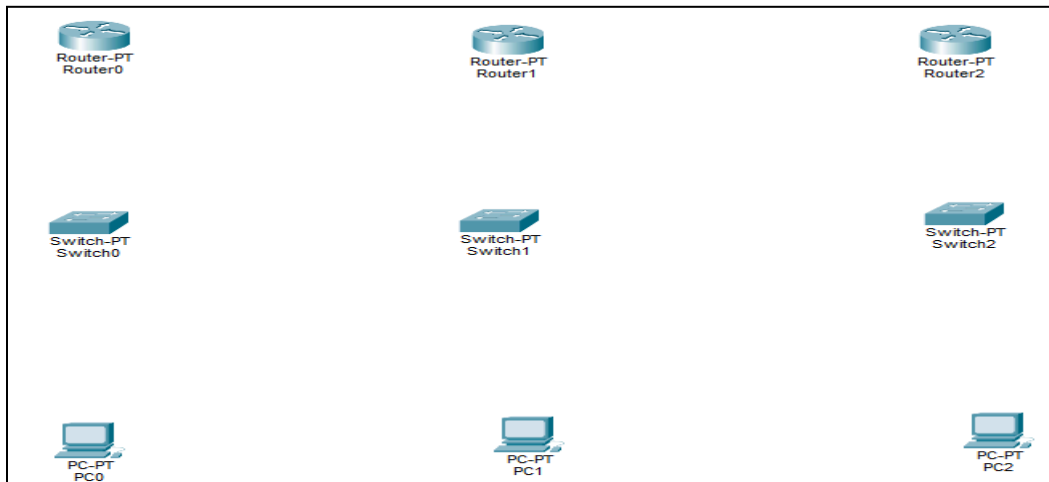
Step 5

From End Devices, Select PCs.



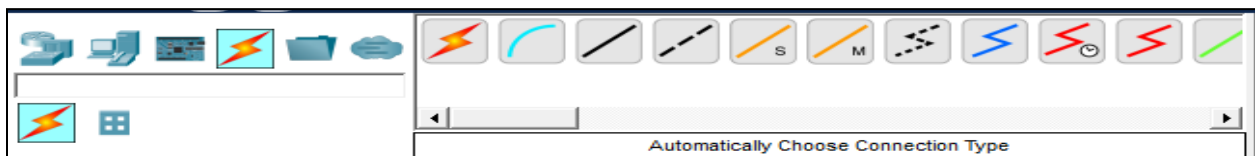
Step 6

Place **minimum three PCs** near each switch.



Step 7

Select the **Connections** (lightning symbol).

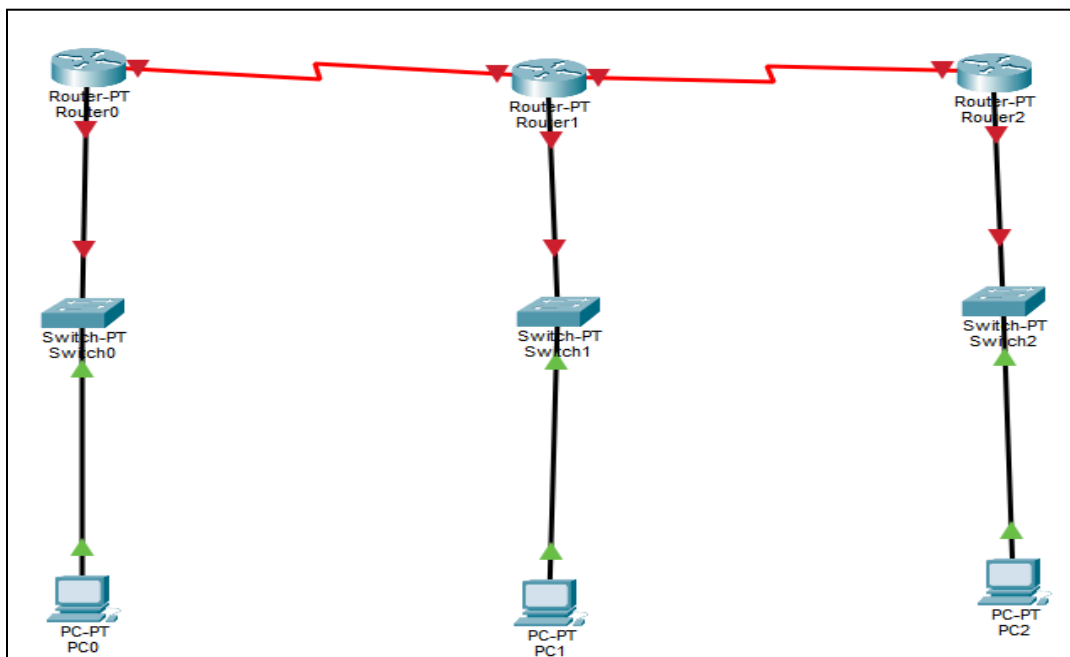


Step 8

Connect PCs to switches using cable

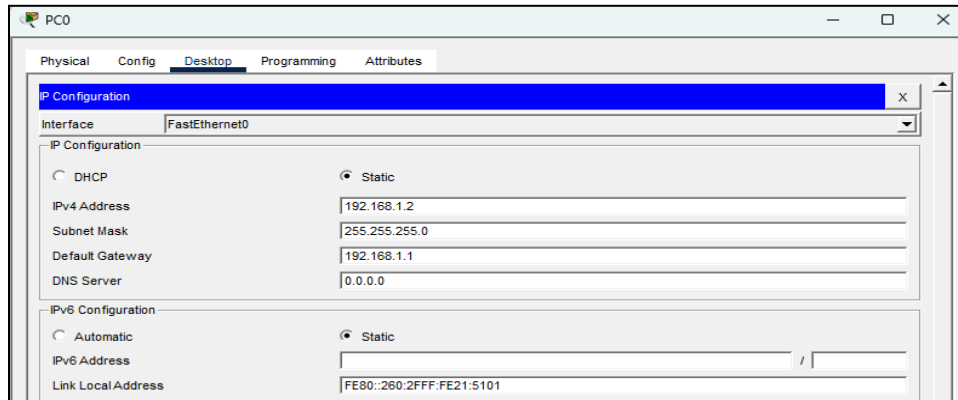
Connect each switch to its respective router using cable.

Connect routers also with each other using cable (Router1–Router2 and Router2–Router3).



Step 9

In the IP configuration window of PC0, set: IP Address, Default Gateway & Subnet Mask



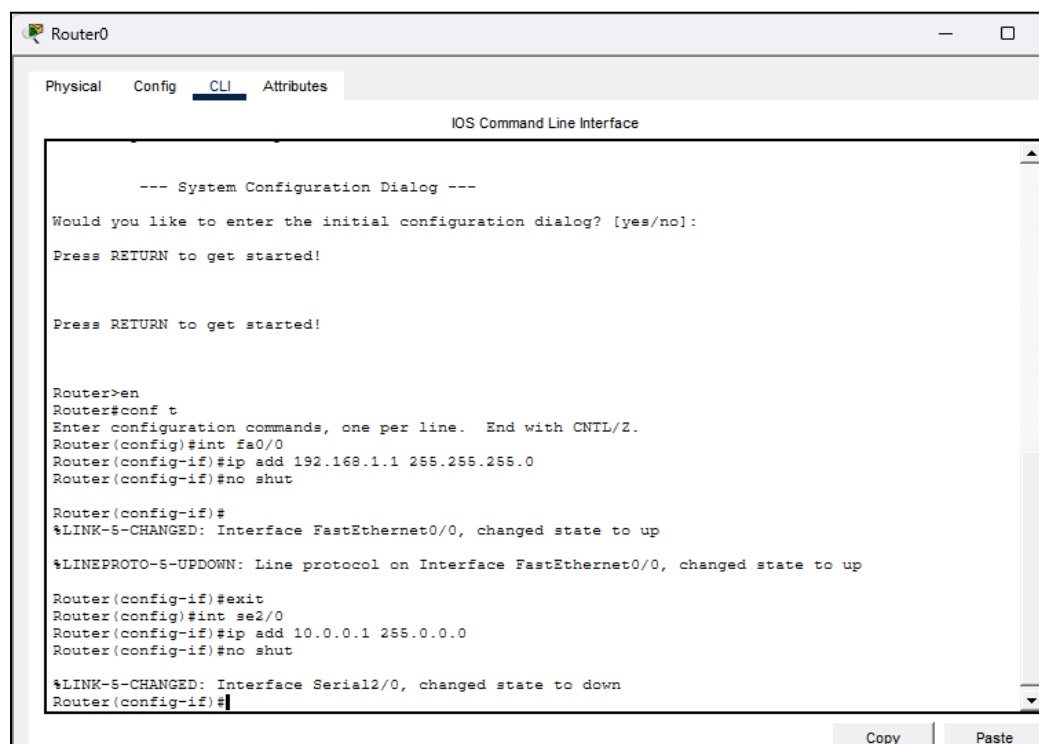
Step 10

Assign new IP address and default Gateway to all the PCs by following same procedure as PC0

	IP Configuration	Default Gateway
PC0	192.168.1.2	192.168.1.1
PC1	192.168.2.2	192.168.2.1
PC2	192.168.3.2	192.168.3.1

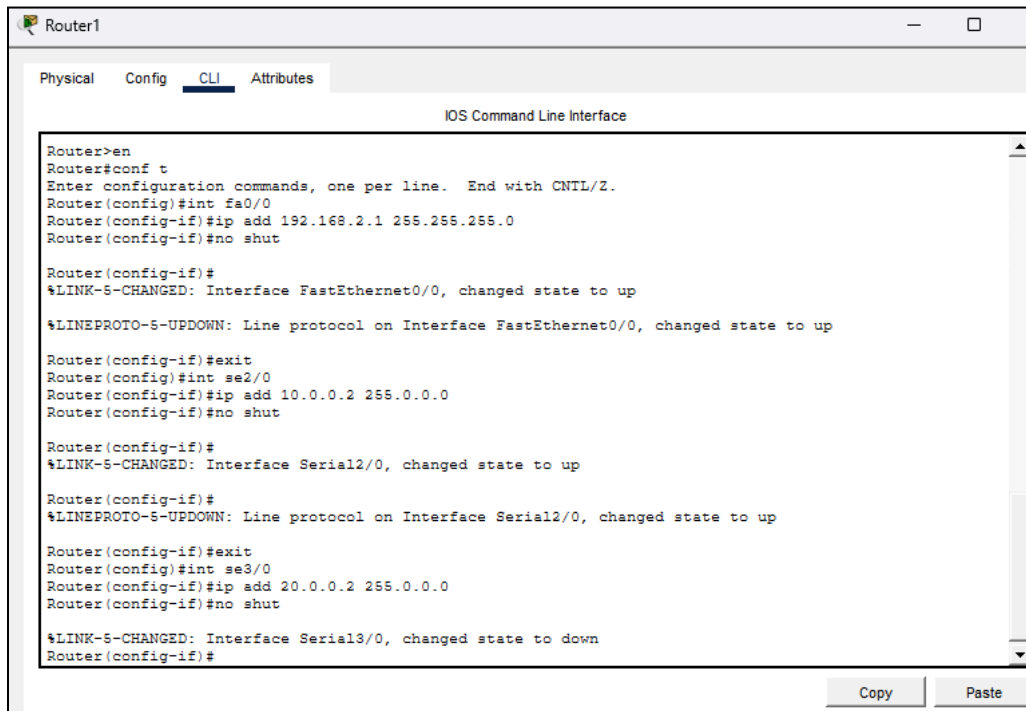
Step 11

Open Router0 → CLI, then configure the **FastEthernet0/0** and **Serial2/0** interfaces by assigning IP addresses and enabling them using the **no shutdown** command.



Step 12

Similarly, configure **Router1** interfaces using the CLI and enable them.

A screenshot of a web-based CLI interface for Router1. The window has tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab shows the IOS Command Line Interface with a scrollable text area containing configuration commands and status messages. At the bottom right are Copy and Paste buttons.

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip add 10.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

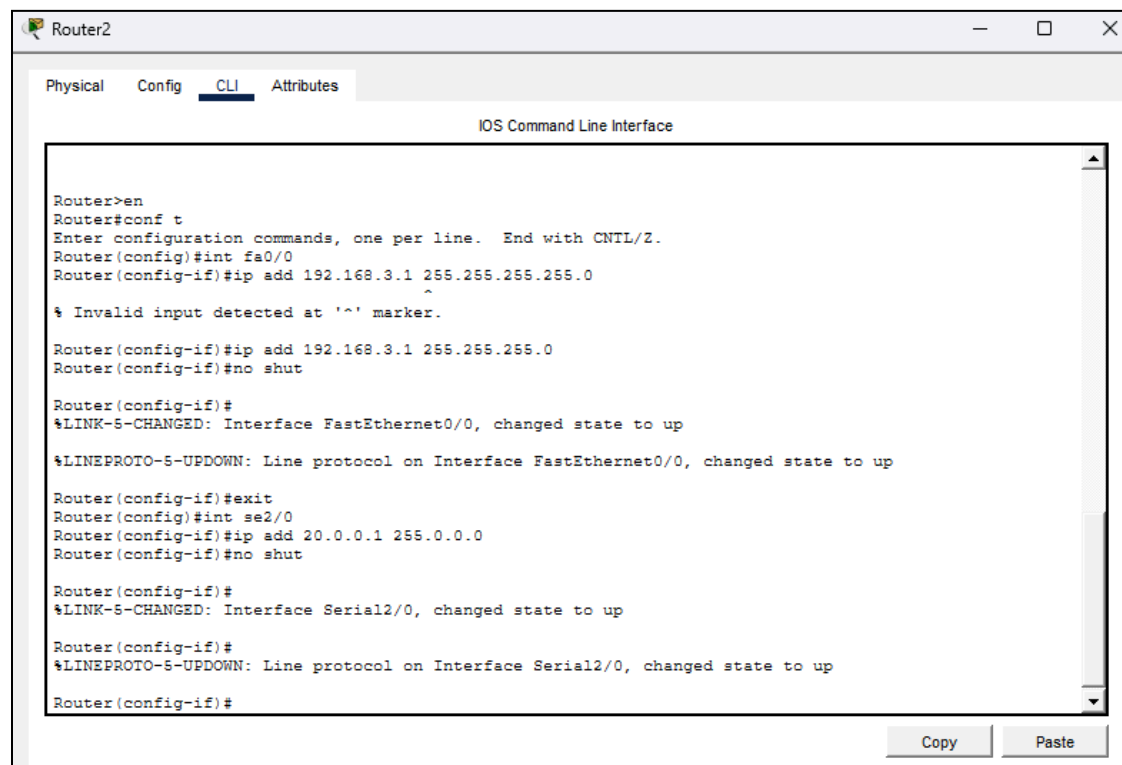
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#int se3/0
Router(config-if)#ip add 20.0.0.2 255.0.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#
```

Step 13

In the same way, configure **Router2** interfaces using the CLI and enable them.

A screenshot of a web-based CLI interface for Router2. The window has tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab shows the IOS Command Line Interface with a scrollable text area containing configuration commands and status messages. At the bottom right are Copy and Paste buttons.

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.3.1 255.255.255.0

% Invalid input detected at '^' marker.

Router(config-if)#ip add 192.168.3.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip add 20.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#
```

Step 14

Open Router0 → CLI and configure OSPF routing by adding all connected networks to Area 0

Router0 – OSPF Configuration (CLI Commands)

```
Router> en
```

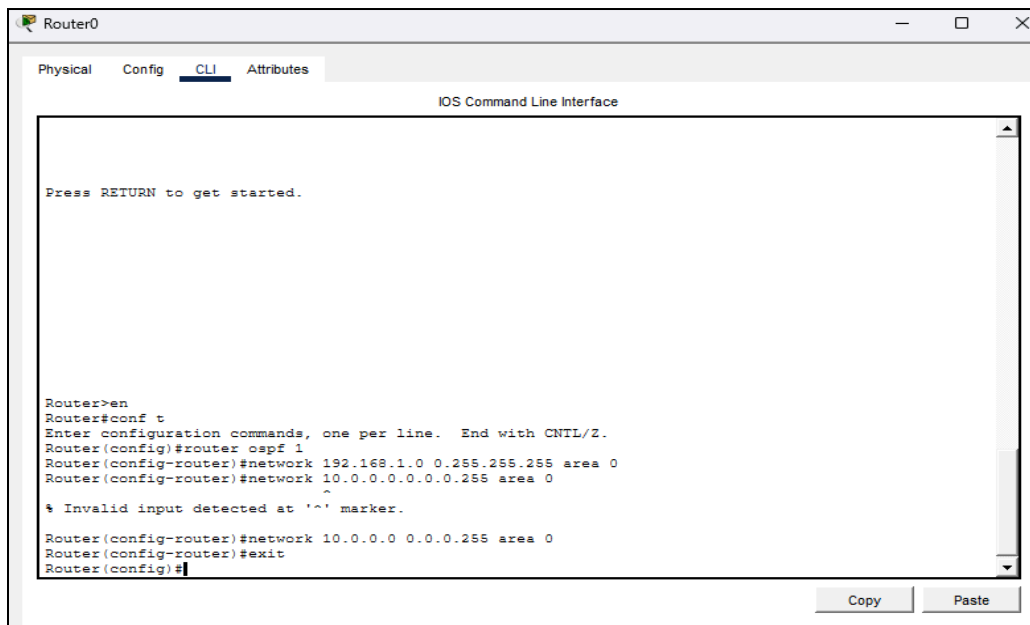
```
Router# conf t
```

```
Router(config)# router ospf 1
```

```
Router(config-router)# network 192.168.1.0 0.255.255.255 area 0
```

```
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

```
Router(config-router)# exit
```



Step 15

OSPF configuration is the same as Router0; only the network IP addresses are different.

Router1– OSPF Configuration (CLI Commands)

```
Router> en
```

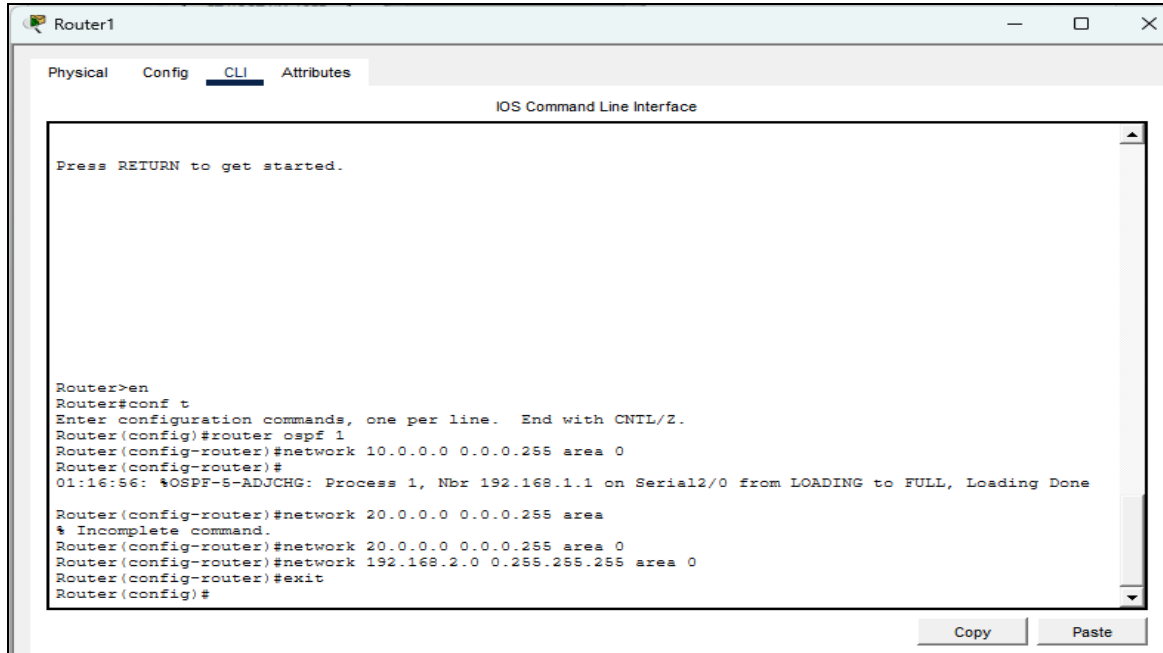
```
Router# conf t
```

```
Router(config)# router ospf 1
```

```
Router(config-router)# network 192.168.2.0 0.255.255.255 area 0
```

```
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

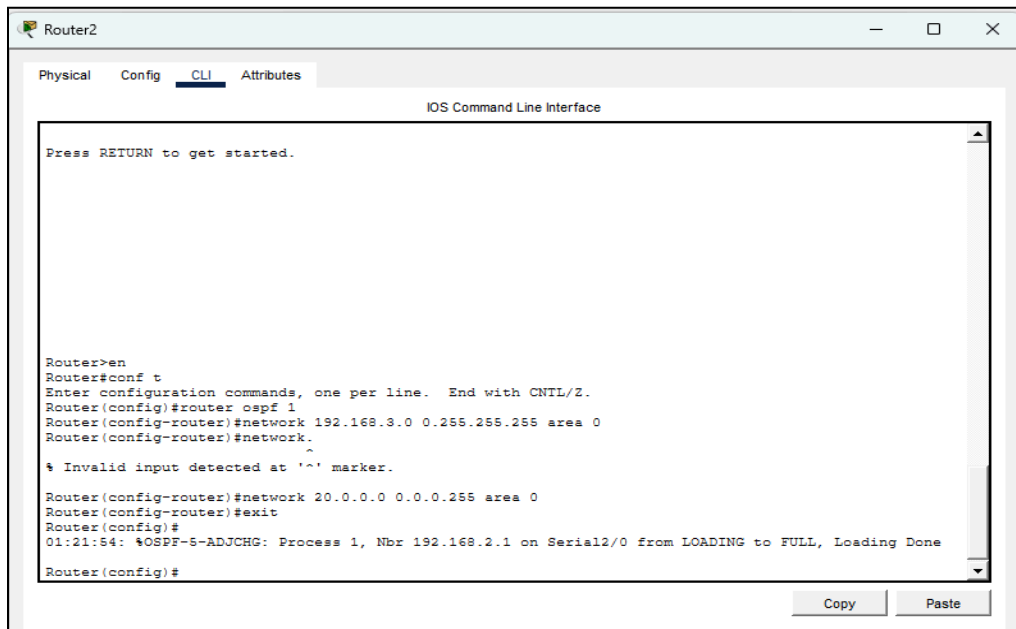
```
Router(config-router)# exit
```



Step 16

In the same way, configure OSPF routing on Router2.

- Router(config-router)# network 192.168.3.0 0.255.255.255 area 0



Step 17

Successful replies confirm that all PCs are connected through OSPF-configured routers and the network is working properly.

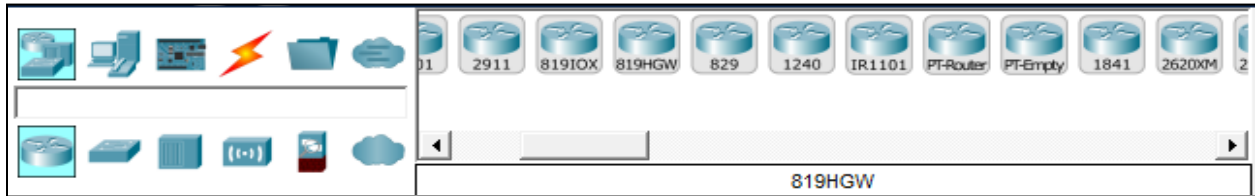
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
...	Successful	PC1	PC2	ICMP		0.000	N	0	(e...	(delete)
...	Successful	PC1	PC0	ICMP		0.000	N	1	(e...	(delete)
...	Successful	PC2	PC0	ICMP		0.000	N	2	(e...	(delete)

Practical No: 7

AIM: Using Packet Tracer, Create a network with three routers with RIPv1 and each router associated network will have minimum three PC. Show Connectivity

Step 1

From Network Devices → Routers, drag 3 routers (R1, R2, R3) to the workspace.



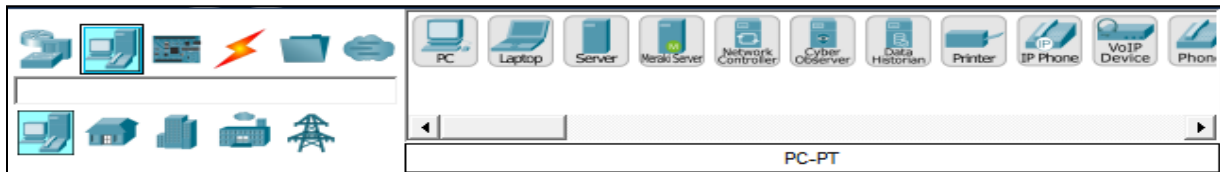
Step 2

Place **three switches**, one near each router.



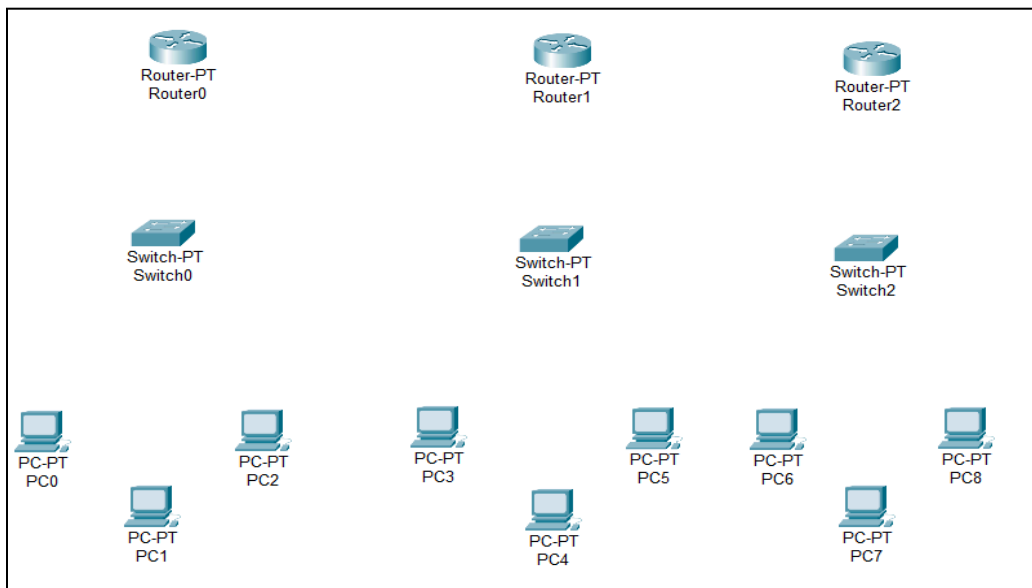
Step 3

From End Devices, drag **9 PCs** (3 PCs near each router).



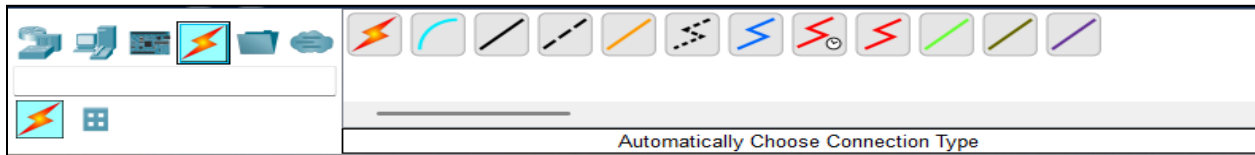
Step 4

Place routers at the top, switches below each router, and three PCs below each switch as shown.



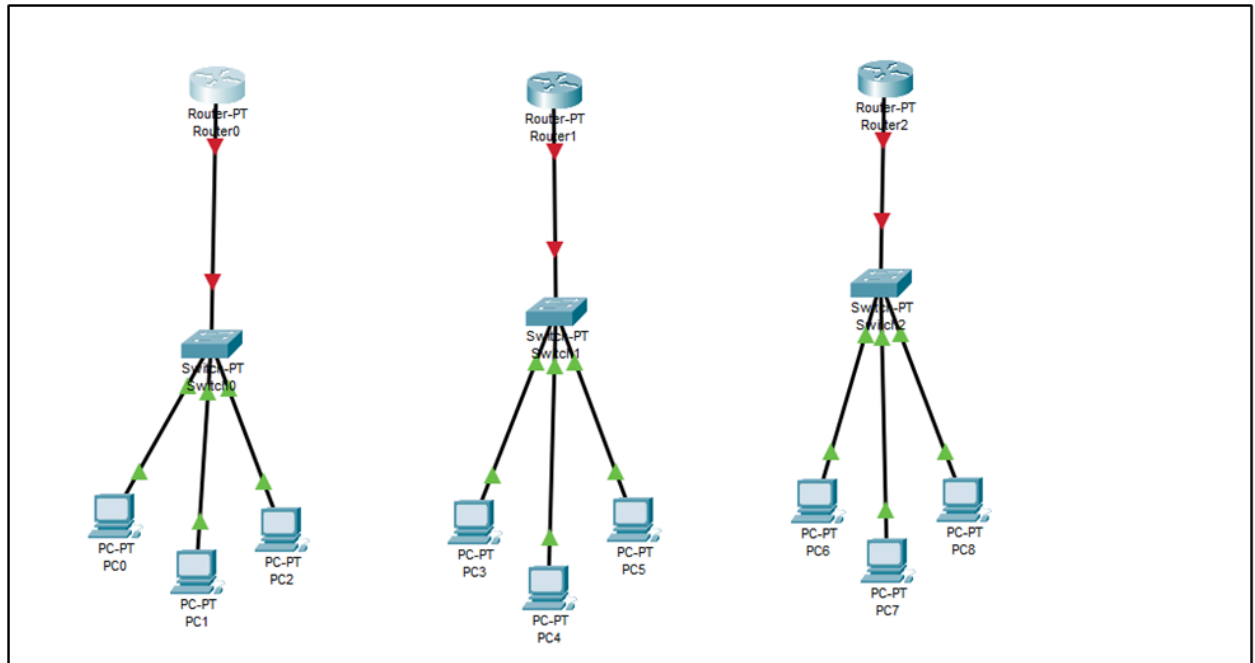
Step 5

From **Connections**, select **cable**.



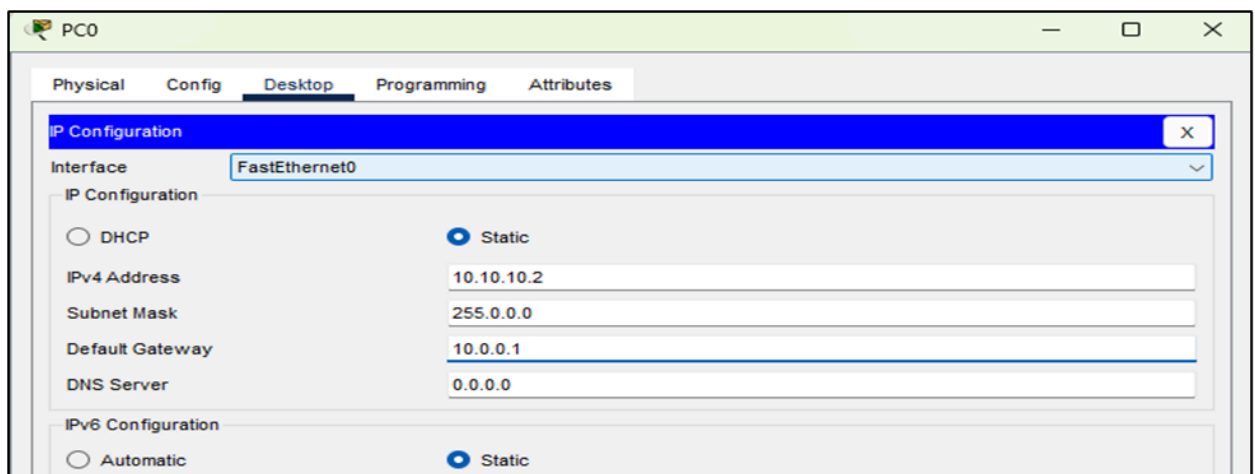
Step 6

Connect PCs to their respective router using FastEthernet ports.



Step 7

In the IP configuration window of PC0, set: IP Address, Default Gateway & Subnet Mask



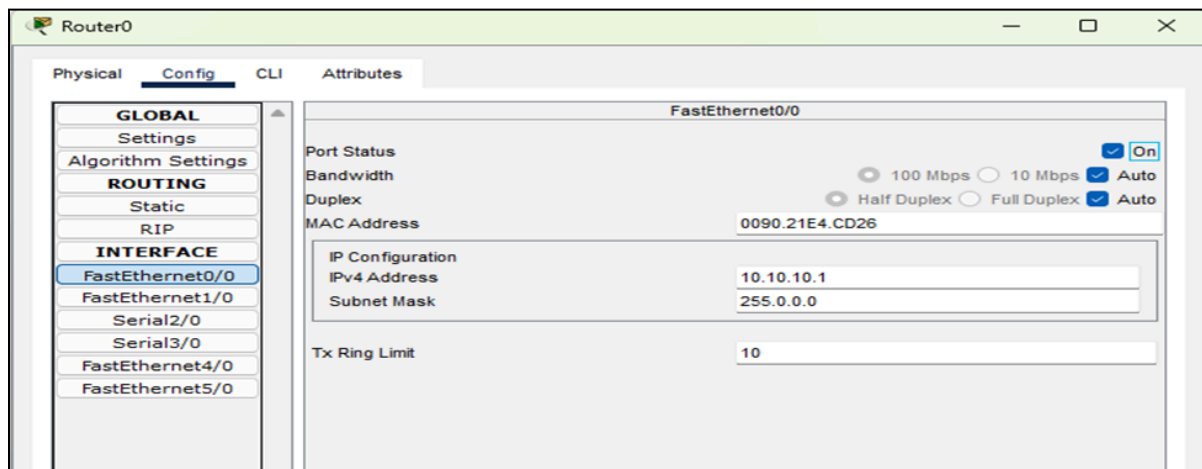
Step 8

Assign new IP address and default Gateway to all the PCs by following same procedure as PC0

	IP Configuration	Default Gateway
PC0	10.10.10.2	10.0.0.1
PC1	10.10.10.3	10.10.10.1
PC2	10.10.10.4	10.10.10.1
PC3	20.20.20.2	20.20.20.1
PC4	20.20.20.3	20.20.20.1
PC5	20.20.20.4	20.20.20.1
PC6	30.30.30.2	30.30.30.1
PC7	30.30.30.3	30.30.30.1
PC8	30.30.30.4	30.30.30.1

Step 9

Click on Router0 → Config → Interfaces and assign IP address and subnet mask.



Step 10

Repeat the steps to all Routers and assign the IP address

	IP Configuration (FastEthernet 0/0)
Router0	10.10.10.1
Router1	20.20.20.1
Router2	30.30.30.1

Step 11

Configure the **Router0 Serial2/0** interface by turning the port **ON** and assigning the IP address and subnet mask

The screenshot shows the 'Router0' configuration window with the 'Config' tab selected. In the left sidebar, under the 'INTERFACE' section, 'Serial2/0' is highlighted. The main panel displays the configuration for 'Serial2/0'. The 'Port Status' is set to 'On' (checked). The 'Duplex' is set to 'Full Duplex'. The 'Clock Rate' is set to '1200'. Under 'IP Configuration', the 'IPv4 Address' is '40.40.40.1' and the 'Subnet Mask' is '255.0.0.0'. The 'Tx Ring Limit' is set to '10'.

Step 12

Repeat the process for assigning new IP Configuration for the rest of the Routers. But in the middle Router you have to **ON** both **Serial2/0** and **Serial3/0** then assign a new IP address for both of them

	IP Configuration
Router0	Serial2/0 40.40.40.1
Router1	Serial2/0 40.40.40.1 Serial3/0 50.50.50.1
Router2	Serial2/0 50.50.50.2

Step 13

Configure **RIP routing on Router0** by opening the **Config** tab, selecting **RIP** under Routing, and adding all the connected **network** addresses one by one.

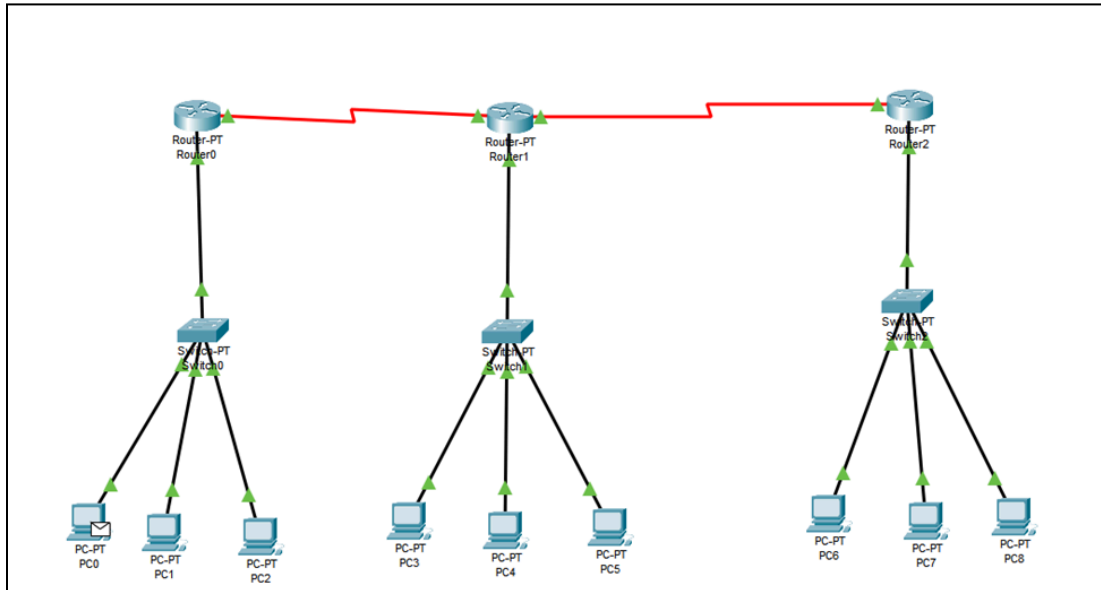
The screenshot shows the 'Router0' configuration window with the 'Config' tab selected. In the left sidebar, under the 'ROUTING' section, 'RIP' is highlighted. The main panel displays the 'RIP Routing' configuration. There is a 'Network' input field with an 'Add' button next to it. Below this, a list of 'Network Address' is shown with the following values: 10.0.0.0, 20.0.0.0, 30.0.0.0, 40.0.0.0, and 50.0.0.0. A 'Remove' button is located at the bottom right of the list.

Step 14

Repeat the same process of configuring **RIP routing** by adding the respective **network addresses** for **Router1** and **Router2**.

Step 15

Send packets between PCs from different router networks.



Step 16

Verify that all pings are **successful**, confirming proper RIP configuration and full network connectivity.

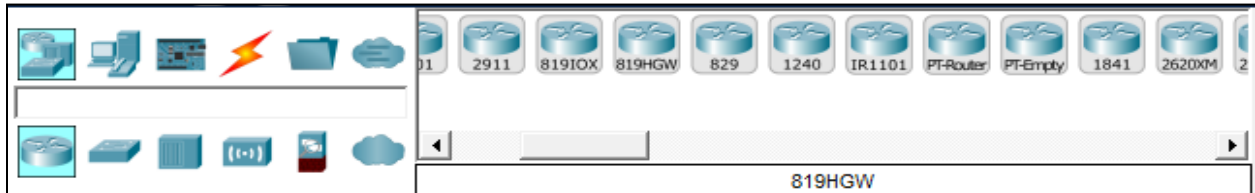
Realtime											
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete	
	Successful	PC4	PC6	ICMP		0.000	N	0	(edit)	(delete)	(delete)
	Successful	PC7	PC4	ICMP		0.000	N	1	(edit)	(delete)	(delete)
	Successful	PC3	PC7	ICMP		0.000	N	2	(edit)	(delete)	(delete)

Practical No: 8

AIM: Using Packet Tracer, Create a network with three routers with RIPv2 and each router associated network will have three PC. Show Connectivity

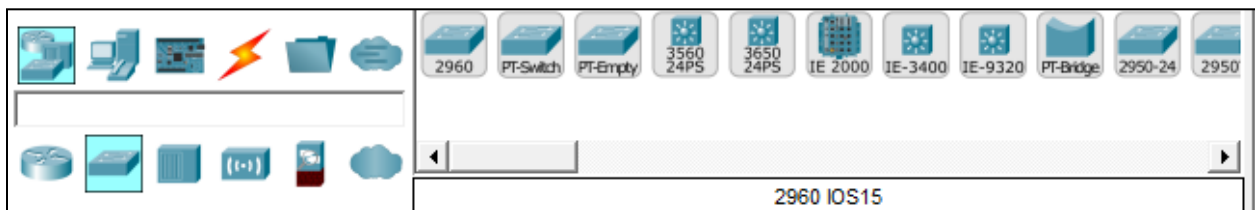
Step 1

Place **three routers** on the workspace and name them **Router0, Router1, and Router2**.



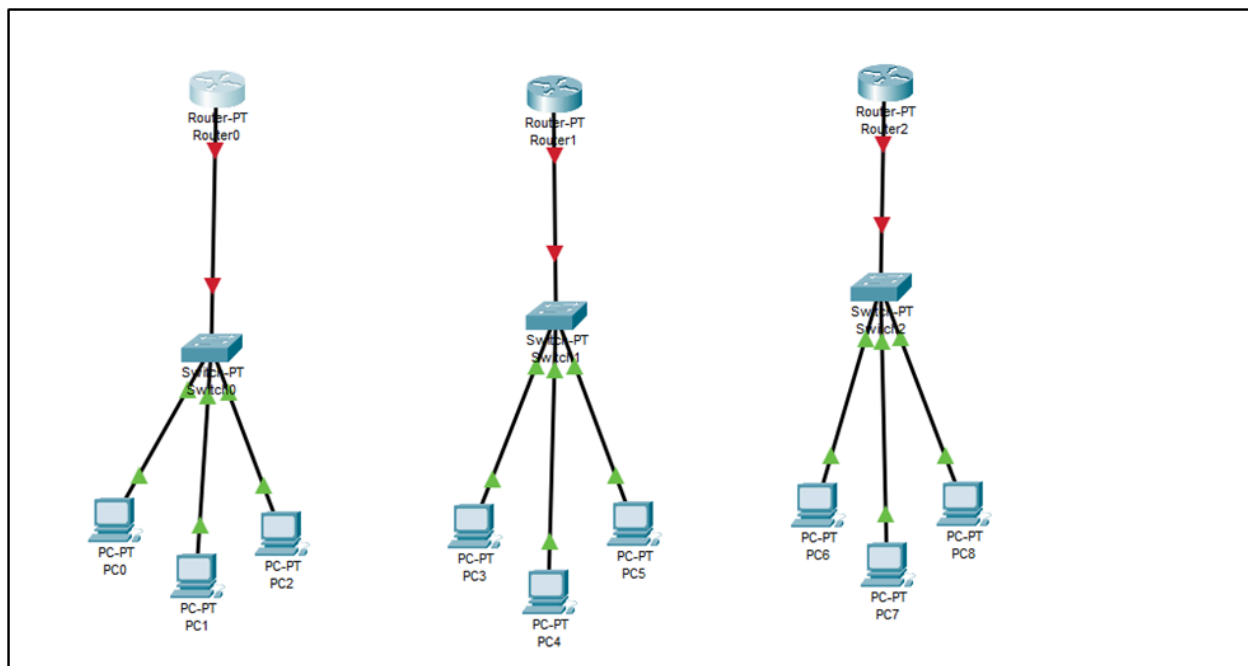
Step 2

Place **three switches**, one for each router network & Place **three PCs for each switch** (total 9 PCs)



Step 3

Connect all PCs to their respective switches, connect each switch to its corresponding router, and connect **Router0 to Router1** and **Router1 to Router2** using serial links.



Step 4

In the IP configuration window of PC0, set: IP Address, Default Gateway & Subnet Mask

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.10.10.2

Subnet Mask: 255.0.0.0

Default Gateway: 10.10.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:43FF:FEE2:7485

Default Gateway:

Step 5

Assign new IP address and default Gateway to all the PCs by following same procedure as PC0

	IP Configuration	Default Gateway
PC0	10.10.10.2	10.10.10.1
PC1	10.10.10.3	10.10.10.1
PC2	10.10.10.4	10.10.10.1
PC3	20.20.20.2	20.20.20.1
PC4	20.20.20.3	20.20.20.1
PC5	20.20.20.4	20.20.20.1
PC6	30.30.30.2	30.30.30.1
PC7	30.30.30.3	30.30.30.1
PC8	30.30.30.4	30.30.30.1

Step 6

Click on Router0 → Config → Interfaces and assign IP address and subnet mask.

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status: ☒ On

Bandwidth: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex: ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address: 0001.97C2.BDAB

IP Configuration

IPv4 Address: 10.10.10.1

Subnet Mask: 255.0.0.0

Tx Ring Limit: 10

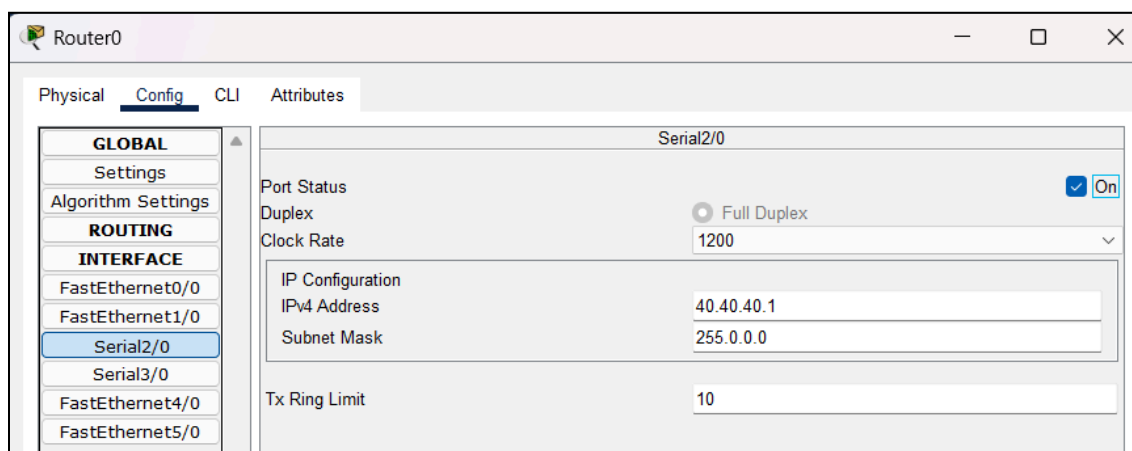
Step 7

Repeat the steps for all Routers and assign the appropriate IP address

	IP Configuration (FastEthernet 0/0)
Router0	10.10.10.1
Router1	20.20.20.1
Router2	30.30.30.1

Step 8

Enable the Serial2/0 interface on Router0 and set its IP address along with the subnet mask.



Step 9

Perform the same IP configuration steps on the other routers. For the middle router, make sure to enable both Serial2/0 and Serial3/0 interfaces, then assign new IP addresses to each.

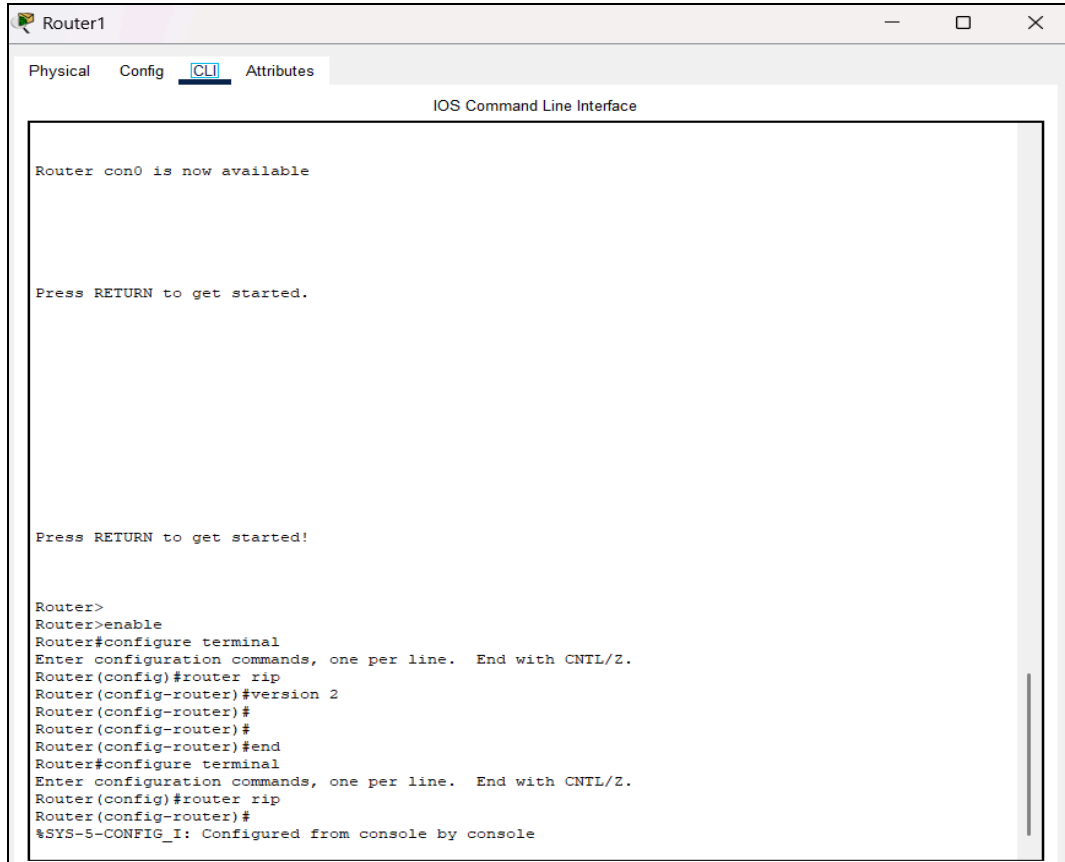
	IP Configuration
Router0	Serial2/0 40.40.40.1
Router1	Serial2/0 40.40.40.2 Serial3/0 50.50.50.1
Router2	Serial2/0 50.50.50.2

Step 10

Go to Router0 → CLI and configure RIP version 2 using the following commands:

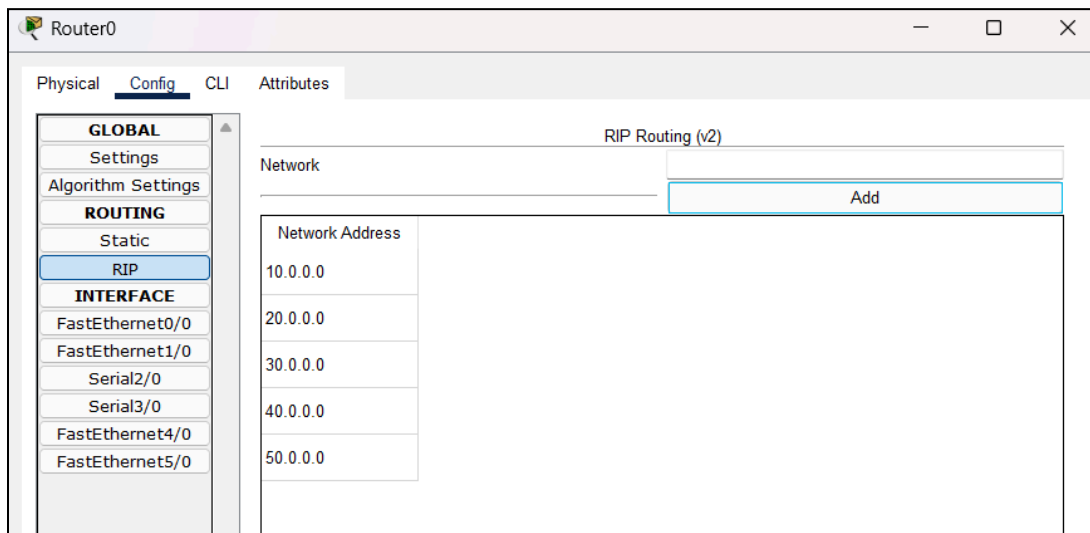
Router0 – RIP Configuration (CLI Commands)

```
Router> en
Router# conf t
Router(config-if)# exit
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# end
```



Step 11

Configure **RIP** routing on Router0 by opening the **Config** tab, selecting **RIP** under Routing, and adding all the connected **network** addresses one by one.

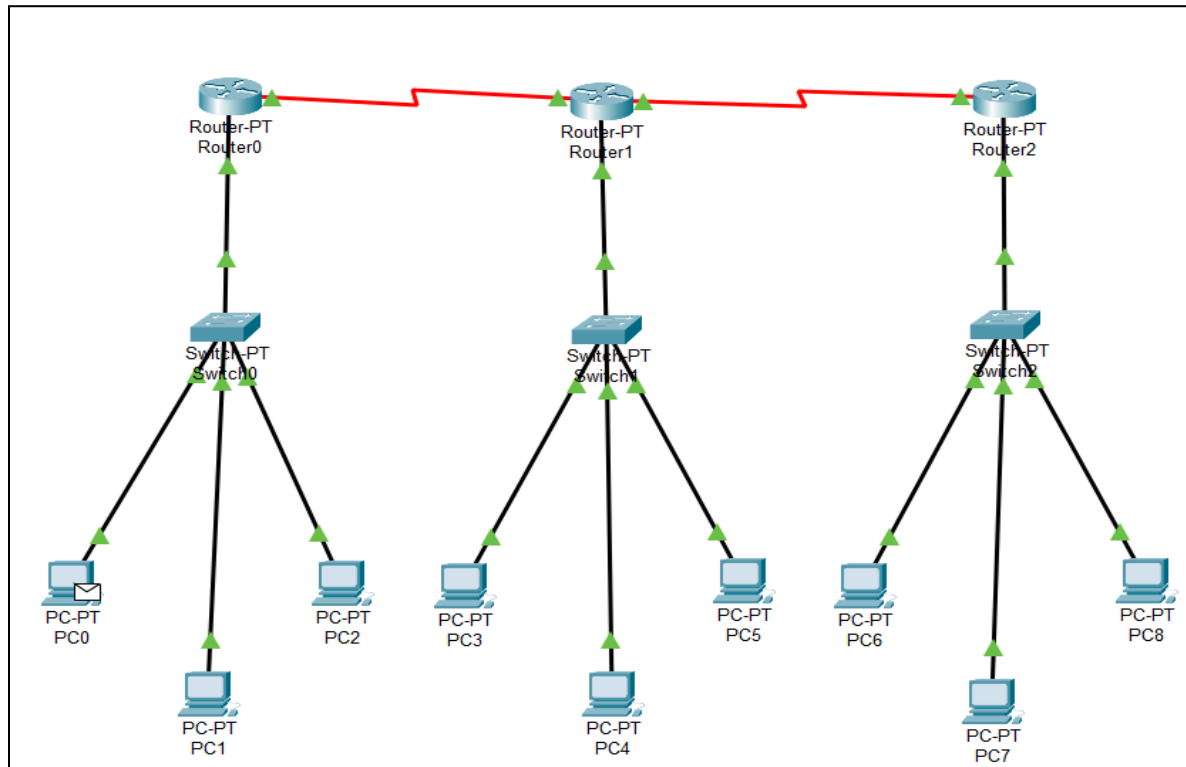


Step 12

Repeat the same process of configuring **RIP routing** by adding the respective **network addresses** for **Router1** and **Router2**.

Step 13

Now, send packets between PCs on different router networks to verify that connectivity and routing are functioning correctly.



Step 14

Ensure that all ping tests succeed, indicating that the RIP routing protocol is operating properly and the network is fully connected.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC2	PC6	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC5	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC5	PC7	ICMP		0.000	N	2	(edit)	(delete)